ENGINEERING ECONOMICS (CE 655)

Chapter 2: Interest and Time Value of Money

Questions & Answers (TU Past Exam Papers)

Q1. What is the time value of money? What are the factors to be considered in calculating the time value of money? [2077 Chaitra, 2080 Chaitra, Multiple Years]

Answer:

Time Value of Money (TVM)

Time Value of Money is the fundamental principle that states that a sum of money available today is worth more than the same sum of money available at a future date. In other words, money has time value because of its potential earning capacity.

Reasons why money has time value:

1. Earning Potential: Money available today can be invested to earn interest or returns, making it worth more than the same amount in the future.
2. Inflation: Over time, inflation reduces the purchasing power of money. A rupee today buys more goods and services than a rupee in the future.
3. Risk and Uncertainty: Future cash flows are uncertain. Present cash is certain. Therefore, people prefer present over future money.
4. Preference for Present Consumption: Human beings generally prefer to consume now rather than later (positive time preference).

Factors to be Considered in Calculating TVM:

1. Present Value (P): The current or present value of a sum of money. It is the starting point of most TVM calculations.
2. Future Value (F): The value of money at some point in the future after earning interest.
3. Interest Rate (i): The rate at which money grows per period. It is the key link between present and future values. Can be simple or compound.
4. Number of Periods (n): The number of time periods (years, months, quarters) over which the money grows or is paid.
5. Periodic Payment (A): A uniform series of equal payments made at regular intervals (annuity).
6. Compounding Frequency (m): How often interest is compounded per year (annually, semi-annually, quarterly, monthly, daily, continuously). Higher compounding frequency results in higher effective interest.
7. Type of Cash Flow: Whether the cash flows are a single amount, uniform series, gradient series, or geometric series — each requires a different formula.
8. Nominal vs. Effective Interest Rate: When compounding is more frequent than annually, nominal and effective rates differ, and the effective rate must be used for accurate calculations.
9. Inflation Rate: For real (inflation-adjusted) analysis, inflation must be considered to distinguish between actual dollar and constant dollar amounts.

Q2. Differentiate between simple and compound interest. What is the relation between nominal and effective interest rate? How long will it take for Rs. 25,000 to triple itself if the interest rate is 8% per year? [2079 Chaitra, 1+2+2 Marks]

Answer:

(A) Simple Interest vs. Compound Interest

Basis	Simple Interest	Compound Interest
Definition	Interest is computed only on the original principal amount	Interest is computed on principal PLUS all previously accumulated interest (interest on interest)
Formula	I = P × i × n, F = P(1 + in)	F = P(1 + i)^n
Interest per Period	Constant (same each period)	Increasing (grows each period)
Application	Short-term loans, simple deposits	Long-term investments, most financial transactions
Practical Use	Simple and rare in modern finance	Standard in banking, investments, loans

(B) Nominal vs. Effective Interest Rate

Nominal Interest Rate (r): The stated or quoted annual interest rate, without considering the effect of compounding within the year.

Effective Interest Rate (i_eff): The actual interest rate earned or paid over a year, accounting for the effect of compounding within the year.

$$i_eff = (1 + r/m)^m - 1$$

Where: r = nominal interest rate per year, m = number of compounding periods per year

For continuous compounding: i_eff = e^r - 1

Example: Nominal rate = 12% per year compounded monthly (m=12):

$$i_eff = (1 + 0.12/12)^{12} - 1 = (1.01)^{12} - 1 = 1.12683 - 1 = 12.683\%$$

The effective rate (12.683%) is higher than the nominal rate (12%) because of monthly compounding.

(C) Time to Triple Rs. 25,000 at 8% per year (Compound Interest)

Given: P = Rs. 25,000, F = 3P = Rs. 75,000, i = 8% per year, n = ?

Using compound interest formula: F = P(1 + i)^n

$$75{,}000 = 25{,}000 \times (1.08)^n$$

$$(1.08)^n = 3$$

$$n \times \ln(1.08) = \ln(3)$$

$$n = \ln(3) / \ln(1.08) = 1.09861 / 0.07696 = 14.27 \text{ years}$$

Therefore, it will take approximately 14.27 years (about 14 years and 3 months) for Rs. 25,000 to triple itself at 8% compound interest per year.

Using the Rule of 72 approximation (for doubling): Years to double = 72/8 = 9 years

Q3. What are the factors to be considered in calculating the time value of money? Distinguish between nominal and effective interest rate with suitable example. Bank 'A' offers 6.25% interest that compounds daily and Bank 'B' offers 6.4% interest that compounds yearly. Which bank do you prefer and why? [2080 Chaitra, 1+2+2 Marks]

Answer:

For factors affecting TVM, refer to Q1 above.

Nominal vs. Effective Interest Rate

Nominal Interest Rate: The stated annual interest rate, not adjusted for intra-year compounding.

Effective Interest Rate: The actual annual rate after accounting for compounding frequency within the year.

```
i_eff = (1 + r/m)^m - 1
```

Bank Comparison: Bank A (6.25% daily compounding) vs Bank B (6.4% annual compounding)

Bank A: r = 6.25% per year, m = 365 (daily compounding)

```
i_eff(A) = (1 + 0.0625/365)^365 - 1
i_eff(A) = (1 + 0.0001712)^365 - 1
i_eff(A) = (1.0001712)^365 - 1 = 1.06449 - 1 = 6.449%
```

Bank B: r = 6.4% per year, m = 1 (annual compounding)

```
i_eff(B) = (1 + 0.064/1)^1 - 1 = 6.4%
```

Comparison:

Bank A effective rate: 6.449% | Bank B effective rate: 6.400%

Conclusion: Bank A (6.449% effective) is preferred over Bank B (6.4% effective) because it offers a higher effective annual interest rate, meaning deposits earn more money over the same period.

Q4. Define time value of money. Calculate effective interest when r = 5% and compounding is: (i) semi-annually, (ii) quarterly, (iii) monthly, and (iv) continuously. [2078 Baishakh, 1+3 Marks]

Answer:

Time value of money refers to the concept that money available today is worth more than the identical sum in the future, because of its potential earning capacity. This core principle of finance holds that, provided money can earn interest, any amount of money is worth more the sooner it is received.

Calculating Effective Interest Rate: r = 5% per year

```
i_eff = (1 + r/m)^m - 1    [for discrete compounding]
i_eff = e^r - 1            [for continuous compounding]
```

(i) Semi-annual compounding (m = 2):

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i_eff = (1 + 0.05/2)^2 - 1 = (1.025)^2 - 1 = 1.050625 - 1 = 5.0625%
```

(ii) Quarterly compounding (m = 4):

```
i_eff = (1 + 0.05/4)^4 - 1 = (1.0125)^4 - 1 = 1.050945 - 1 = 5.0945%
```

(iii) Monthly compounding (m = 12):

```
i_eff = (1 + 0.05/12)^12 - 1 = (1.004167)^12 - 1 = 1.05116 - 1 = 5.116%
```

(iv) Continuous compounding (m -> infinity):

```
i_eff = e^0.05 - 1 = 1.05127 - 1 = 5.127%
```

Compounding	m	Effective Rate
Semi-annual	2	5.0625%
Quarterly	4	5.0945%
Monthly	12	5.116%
Continuous	∞	5.127%

Observation: As compounding frequency increases, the effective interest rate increases but approaches a limit (continuous compounding). For a given nominal rate, continuous compounding gives the maximum effective rate.

Q5. Explain different types of standard cash flow with their economic equivalence formulae. [2079 Ashwin, 4 Marks]

Answer:

In engineering economics, cash flows are classified into five standard types, each with associated interest factor formulas:

Type 1: Single Cash Flow

A single amount P at the present time or F at a future time.

Formulas:

```
F = P(F/P, i%, n) = P(1+i)^n          [Single Payment Compound Amount]
P = F(P/F, i%, n) = F/(1+i)^n          [Single Payment Present Worth]
```

(F/P, i%, n) is called the Single Payment Compound Amount Factor

(P/F, i%, n) is called the Single Payment Present Worth Factor

Type 2: Uniform Series (Equal Payment / Annuity)

A series of equal payments A made at the end of each period for n periods.

```
F = A(F/A, i%, n) = A[(1+i)^n - 1]/i    [Uniform Series Compound Amount]
A = F(A/F, i%, n) = Fi/[(1+i)^n - 1]    [Sinking Fund Factor]
P = A(P/A, i%, n) = A[(1+i)^n - 1]/[i(1+i)^n]   [Uniform Series PW]
A = P(A/P, i%, n) = Pi(1+i)^n/[(1+i)^n - 1]     [Capital Recovery Factor]
```

Type 3: Uneven Cash Flow Series

A series of different cash flows at different time periods. There is no standard factor formula; each cash flow must be converted to its present or future worth individually and then summed.

```
P = Sum[F_t / (1+i)^t]  for t = 1 to n
```

Type 4: Linear Gradient Series (Arithmetic Gradient)

A series of cash flows that change by a constant amount G per period. The first cash flow is A1 at end of period 1, and each subsequent payment increases (or decreases) by G.

```
P = A1(P/A, i, n) + G(P/G, i, n)
P/G factor = [(1+i)^n - in - 1] / [i^2 (1+i)^n]
A = A1 + G(A/G, i, n)   where  A/G = 1/i - n/[(1+i)^n - 1]
```

Type 5: Geometric Gradient Series

A series of cash flows that change by a constant percentage g per period.

$$P = A1 \times (P/A1, g, i, n)$$

When g ≠ i:

$$P = A1 \times [1 - (1+g)^n(1+i)^{(-n)}] / (i - g)$$

When g = i:

$$P = A1 \times n / (1+i)$$

Type	Cash Flow Pattern	Key Factor
Single CF	One lump sum at t=0 or t=n	F/P or P/F
Uniform Series	Equal payments each period	P/A, A/P, F/A, A/F
Uneven Series	Irregular payments	Sum of individual P/F
Arithmetic Gradient	Changes by constant G each period	P/G, A/G
Geometric Gradient	Changes by constant % each period	P/A1,g,i,n

Q6. Explain continuous compounding. What is the future equivalent of a continuous funds flow of $10,000 per year for N=12 years at 20% compounding continuously? [2076 Bhadra, 4 Marks]

Answer:

Continuous Compounding

Continuous compounding refers to the theoretical limit of compounding frequency where interest is compounded infinitely many times per year. In practice, it serves as an approximation for very frequent compounding (e.g., daily or sub-daily).

When interest is compounded m times per year at nominal rate r, the effective annual rate is:

$$i_eff = (1 + r/m)^m - 1$$

As m → ∞ (continuous compounding):

$$i_eff = e^r - 1$$

For continuous compounding, the standard formulas become:

$$F = P \times e^{(rn)} \quad \text{[Single payment, continuous compounding]}$$

$$P = F \times e^{(-rn)} \quad \text{[Single payment, continuous discounting]}$$

For Continuous Funds Flow (Continuous Uniform Series):

When cash flows occur continuously at a constant rate A_bar per year, the future equivalent is:

$$F = A_bar \times [(e^{(rn)} - 1) / r]$$

This formula is derived by integrating continuous cash flows with continuous compounding.

Calculation: A_bar = $10,000/year, r = 20% = 0.20, n = 12 years

$$F = 10{,}000 \times [(e^{(0.20 \times 12)} - 1) / 0.20]$$

$$F = 10{,}000 \times [(e^{2.4} - 1) / 0.20]$$

$$e^{2.4} = 11.0232$$

$$F = 10{,}000 \times [(11.0232 - 1) / 0.20]$$

$$F = 10{,}000 \times [10.0232 / 0.20]$$

$$F = 10{,}000 \times 50.116$$

$$F = \$501{,}160$$

The future equivalent of the continuous fund flow is $501,160.

Q7. A bank is starting its nominal interest rate at 9% p.a. and compounding quarterly. Calculate the effective interest rate for (i) a year (ii) a quarter (iii) a month (iv) half year. [2075 Bhadra, 4 Marks]

Answer:

Given: Nominal interest rate r = 9% per year, Compounding: Quarterly (m = 4)

Step 1: Determine effective rate per compounding period

$$\text{Effective rate per quarter} = r/m = 9\%/4 = 2.25\% \text{ per quarter}$$

(i) Effective Annual Rate (per year):

$$i_year = (1 + r/m)^m - 1 = (1 + 0.09/4)^4 - 1$$

$$i_year = (1.0225)^4 - 1 = 1.09308 - 1 = 0.09308 = 9.308\%$$

(ii) Effective Rate per Quarter:

$$i_quarter = r/m = 0.09/4 = 0.0225 = 2.25\%$$

(iii) Effective Rate per Month:

There are 3 months per quarter, so from the quarterly effective rate:

$$i_month = (1 + i_quarter)^{(1/3)} - 1 = (1.0225)^{(1/3)} - 1$$

$$i_month = 1.007444 - 1 = 0.7444\%$$

(iv) Effective Rate per Half Year (semi-annual):

There are 2 quarters per half year:

$$i_half = (1 + i_quarter)^2 - 1 = (1.0225)^2 - 1$$

$$i_half = 1.04551 - 1 = 0.04551 = 4.551\%$$

Period	Formula	Effective Rate
Annual	(1.0225)^4 - 1	9.308%
Semi-annual (Half year)	(1.0225)^2 - 1	4.551%
Quarterly	r/m = 9%/4	2.250%
Monthly	(1.0225)^(1/3) - 1	0.744%

Q8. If you deposit Rs. 5,000 at the end of each month into a bank account that offers interest rate of 12% compounded monthly for 5 years, after that what is the maximum fixed amount you can withdraw each month indefinitely while maintaining the principal balance? [2082 Kartik, Q3a, 6 Marks]

Answer:

Given:

- Monthly deposit: A = Rs. 5,000

- Interest rate: 12% compounded monthly => i = 12%/12 = 1% per month
- Period of deposit: 5 years = 60 months
- After 5 years: withdraw indefinitely (perpetuity)

Step 1: Find the Future Value (FV) after 5 years of deposits

$$FV = A \times (F/A, 1\%, 60) = A \times [(1+i)^n - 1] / i$$

$$FV = 5{,}000 \times [(1.01)^{60} - 1] / 0.01$$

$$(1.01)^{60} = 1.81670$$

$$FV = 5{,}000 \times [1.81670 - 1] / 0.01 = 5{,}000 \times 81.6697 = \text{Rs. } 4{,}08{,}348.40$$

Step 2: Maximum perpetual withdrawal (withdrawing only the interest each month)

To withdraw indefinitely while maintaining the principal, only interest earned each month can be withdrawn:

$$W = FV \times i = 4{,}08{,}348.40 \times 0.01$$

$$W = \text{Rs. } 4{,}083.48 \text{ per month}$$

The maximum fixed amount that can be withdrawn each month indefinitely while maintaining the principal is Rs. 4,083.48.

Q9. How much money should be deposited each year for 12 years if you wish to withdraw Rs. 3,000 each year for 5 years beginning at the end of 14th year? Take i = 9% compounded quarterly. [2082 Kartik, Q1b, 4 Marks]

Answer:

Given:

- Annual deposit A1 for 12 years (years 1 to 12)
- Withdrawals: Rs. 3,000 per year for 5 years starting at end of year 14 (i.e., at years 14, 15, 16, 17, 18)
- Nominal interest rate i = 9% per year compounded quarterly

Step 1: Find Effective Annual Interest Rate

$$i_{eff} = (1 + 0.09/4)^4 - 1 = (1.0225)^4 - 1 = 9.308\%$$

Let i = 9.308% per year for annual calculations.

Step 2: Find Present Worth of Withdrawals at year 13

The 5 annual withdrawals of Rs. 3,000 start at end of year 14. First, find PW at end of year 13 (one period before first withdrawal):

$$PW_{13} = 3{,}000 \times (P/A, 9.308\%, 5)$$

$$(P/A, 9.308\%, 5) = [(1.09308)^5 - 1] / [0.09308 \times (1.09308)^5]$$

$$(1.09308)^5 = 1.56145$$

$$(P/A, 9.308\%, 5) = [1.56145 - 1] / [0.09308 \times 1.56145] = 0.56145 / 0.14534 = 3.8630$$

$$PW_{13} = 3{,}000 \times 3.8630 = \text{Rs. } 11{,}589$$

Step 3: Find Present Worth at year 0 (t = 0)

$$PW_0 = PW_{13} \times (P/F, 9.308\%, 13) = 11{,}589 \times (1.09308)^{(-13)}$$

$$(1.09308)^{13} = 3.2534$$

$$PW_0 = 11{,}589 / 3.2534 = Rs.\ 3{,}562.2$$

Step 4: Find annual deposit A1 for 12 years

$$PW_0 = A1 \times (P/A,\ 9.308\%,\ 12)$$

$$(P/A,\ 9.308\%,\ 12) = [(1.09308)^{12} - 1] / [0.09308 \times (1.09308)^{12}]$$

$$(1.09308)^{12} = 2.9764$$

$$(P/A,\ 9.308\%,\ 12) = [2.9764 - 1] / [0.09308 \times 2.9764] = 1.9764 / 0.27699 = 7.1347$$

$$A1 = PW_0 / (P/A,\ 9.308\%,\ 12) = 3{,}562.2 / 7.1347 = Rs.\ 499.3$$

Annual deposit required: approximately Rs. 499 per year for 12 years.

Q10. Assume you contribute Rs. 5,000 monthly to a retirement account that earns an interest rate of 10%, compounded quarterly. Calculate the total balance after 12 years, considering that interest is applied to the deposits made during each compounding period. [2081 Chaitra, Q1b, 4 Marks]

Answer:

Given:

- Monthly deposit: A = Rs. 5,000
- Nominal interest rate: r = 10% per year, compounded quarterly (m = 4)
- Period: n = 12 years = 48 quarters = 144 months

Step 1: Find effective monthly interest rate

Nominal rate per quarter = 10%/4 = 2.5% per quarter

$$\text{Effective monthly rate } (i_m) = (1 + 0.025)^{(1/3)} - 1 = (1.025)^{(0.3333)} - 1$$

$$i_m = 1.008265 - 1 = 0.8265\% \text{ per month}$$

Step 2: Calculate Future Value of monthly deposits over 12 years

$$FV = A \times (F/A,\ i_m,\ n) = 5{,}000 \times [(1 + i_m)^{144} - 1] / i_m$$

$$(1.008265)^{144} = (1.008265)^{144}$$

Note: (1.008265)^144 = (1 + effective quarterly rate)^48 = (1.025)^48

$$(1.025)^{48} = 3.27149$$

$$FV = 5{,}000 \times [(3.27149 - 1) / 0.008265]$$

$$FV = 5{,}000 \times [2.27149 / 0.008265]$$

$$FV = 5{,}000 \times 274.84$$

$$FV = Rs.\ 13{,}74{,}200$$

Total balance after 12 years of Rs. 5,000 monthly contributions at 10% compounded quarterly: approximately Rs. 13,74,200.

Q11. An individual makes 5 annual deposits of Rs. 2,000 in a saving account that pays interest rate of 4% per year. One year after making the last deposit, the interest rate changes to 6% per year. Five years after the last deposit, the

accumulated money is withdrawn from the account. How much money is withdrawn? [2081 Chaitra, Q2b, 6 Marks]

Answer:

Given:

- 5 annual deposits of Rs. 2,000 (at end of years 1 through 5)
- Interest rate: 4% per year during deposit years (year 1 to 5)
- Interest rate changes to 6% per year one year after last deposit (i.e., after year 5, for years 6 to 10)
- Withdrawal at end of year 10 (5 years after last deposit)

Step 1: Find Future Value of 5 deposits at end of year 5 (at 4%)

$$FV_5 = 2{,}000 \times (F/A,\ 4\%,\ 5) = 2{,}000 \times [(1.04)^5 - 1] / 0.04$$

$$(1.04)^5 = 1.21665$$

$$FV_5 = 2{,}000 \times [1.21665 - 1] / 0.04 = 2{,}000 \times 5.4163 = \text{Rs. } 10{,}832.60$$

Step 2: Advance FV_5 by 5 more years at 6% to year 10

$$FV_10 = FV_5 \times (F/P,\ 6\%,\ 5) = 10{,}832.60 \times (1.06)^5$$

$$(1.06)^5 = 1.33823$$

$$FV_10 = 10{,}832.60 \times 1.33823 = \text{Rs. } 14{,}497.95$$

Amount withdrawn at end of year 10: Rs. 14,497.95 (approximately Rs. 14,498).

Q12. The two cash flow transactions shown below are said to be equivalent at 10% interest, compounded annually. Find the unknown X value which satisfies the equivalence. [2081 Chaitra Q3a, 2076 Baisakh Q2a, 6 Marks]

Answer:

Cash Flow Diagram Description:

Left side (original): Rs. 200 at t=5, Rs. 150 at t=3, Rs. 150 at t=2, Rs. 100 at t=1, Rs. 100 at t=2 (upward arrows), and an initial outflow at t=0.

Note: Based on the cash flow diagram from the exam paper (2081 Chaitra Q3a):

Left side: Rs. 200 at t=5, Rs. 200 at t=1, Rs. 150 at t=2, Rs. 150 at t=3, Rs. 100 at t=2, Rs. 100 at t=1

Right side: Equal payments of X at t=0,1,2,3,4,5

Standard Problem Setup (Typical Form):

Left side cash flows: Rs. 100 at t=1, Rs. 100+Rs. 100=Rs. 200 at t=2, Rs. 150 at t=3, Rs. 150 at t=4, Rs. 200 at t=5

Right side: X at t=1 through t=5 (uniform series)

Step 1: Find PW of left-side cash flows at 10%:

$$PW_L = 100(P/F,10\%,1) + 200(P/F,10\%,2) + 150(P/F,10\%,3) + 150(P/F,10\%,4) + 200(P/F,10\%,5)$$

$$(P/F,10\%,1) = 0.9091,\ (P/F,10\%,2) = 0.8264,\ (P/F,10\%,3) = 0.7513$$

$$(P/F,10\%,4) = 0.6830,\ (P/F,10\%,5) = 0.6209$$

$$PW_L = 100(0.9091) + 200(0.8264) + 150(0.7513) + 150(0.6830) + 200(0.6209)$$

$$PW_L = 90.91 + 165.28 + 112.70 + 102.45 + 124.18 = Rs.\ 595.52$$

Step 2: Set equal to PW of uniform series X for 5 years:

$$PW_R = X \times (P/A,\ 10\%,\ 5) = X \times 3.7908$$

$$595.52 = X \times 3.7908$$

$$X = 595.52 / 3.7908 = Rs.\ 157.09$$

The unknown value X = Rs. 157.09 per period.

Note: The exact X value depends on the specific cash flow diagram. The above uses one common variant of this problem. For the exam diagram from 2076 Baisakh and 2081 Chaitra, carefully read the diagram and apply the same approach.

Q13. Mr. Hari, father of Ram, had deposited Rs. 2,50,000 in a bank 10 years ago at an interest of 10% pa compounded quarterly. After knowing this, Ram is planning to deposit Rs. 50,000 at the end of this year and planned to increase the deposit annually by 15% till 5 years' end but with revised interest scheme of 9.5% interest compounding monthly. What would be the accumulated cash in a bank at the end of 10th year of Ram's first deposit? [2081 Ashwin, Q2b, 6 Marks]

Answer:

This problem has two parts:

Part A: Find FV of Hari's Rs. 2,50,000 after 10+10 = 20 years? NO — Hari deposited 10 years ago. At the time of Ram's planning (now = t=0), Hari's account has been growing for 10 years. We need to find the current value of Hari's deposit, then advance it 10 more years.

Wait — let me re-read: Ram's question asks about 'accumulated cash in a bank at the end of 10th year of Ram's first deposit'. So we need value at end of year 10 from now.

Part A: Hari's Rs. 2,50,000 deposited 10 years ago at 10% compounded quarterly

Current value of Hari's deposit (now, at t=0 for Ram):

$$FV_Hari_now = 2,50,000 \times (1 + 0.10/4)^{(4\times10)} = 2,50,000 \times (1.025)^{40}$$

$$(1.025)^{40} = 2.68506$$

$$FV_Hari_now = 2,50,000 \times 2.68506 = Rs.\ 6,71,265$$

Advance this to year 10 from now (at 10% compounded quarterly, continuing):

$$FV_Hari_10 = 6,71,265 \times (1.025)^{40} = 6,71,265 \times 2.68506 = Rs.\ 18,02,015$$

Part B: Ram's geometric gradient deposits for 5 years (A1 = 50,000, g = 15%, r = 9.5% monthly)

Effective annual rate for Ram's account (9.5% monthly => 9.5% compounded monthly):

$$i_eff = (1 + 0.095/12)^{12} - 1 = (1.007917)^{12} - 1 = 1.09921 - 1 = 9.921\%$$

Geometric gradient PW (deposits at end of years 1-5, g = 15%, i_eff = 9.921%):

$$PW_Ram = A1 \times [1 - (1+g)^n(1+i)^{(-n)}] / (i - g)$$

$$PW_Ram = 50,000 \times [1 - (1.15)^5 \times (1.09921)^{(-5)}] / (0.09921 - 0.15)$$

$$(1.15)^5 = 2.01136, \quad (1.09921)^5 = 1.60075$$

$$PW_Ram = 50,000 \times [1 - 2.01136/1.60075] / (-0.05079)$$

$$PW_Ram = 50,000 \times [1 - 1.25654] / (-0.05079)$$

```
PW_Ram = 50,000 × [-0.25654] / (-0.05079) = 50,000 × 5.0508 = Rs. 2,52,540
```

FV of Ram's deposits at end of year 10 (advance PW by 10 years at 9.921%):

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FV_Ram_10 = 2,52,540 × (1.09921)^10 = 2,52,540 × 2.56239 = Rs. 6,47,090
```

Total accumulated cash at end of year 10:

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Total = FV_Hari_10 + FV_Ram_10 = Rs. 18,02,015 + Rs. 6,47,090 = Rs.
24,49,105
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Total accumulated cash at end of 10th year of Ram's first deposit: approximately Rs. 24,49,105.

Q14. Differentiate the nominal and effective interest with the support of 10% bank's interest which compounds daily. [2081 Ashwin, Q2a, 2+2 Marks]

Answer:

Nominal Interest Rate

The nominal interest rate (r) is the stated or quoted annual interest rate, without accounting for the effect of intra-year compounding. It is also called the Annual Percentage Rate (APR). It does not represent the true cost or yield of a financial instrument if compounding occurs more than once a year.

Effective Interest Rate

The effective interest rate (i_eff) is the actual annual interest rate that accounts for the effect of compounding frequency within the year. It represents the true annual cost or yield of an investment or loan.

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i_eff = (1 + r/m)^m - 1
```

where r = nominal rate, m = compounding periods per year

Example: 10% nominal rate, compounded daily (m = 365)

```
r = 10% = 0.10 per year
m = 365 (daily compounding)
i_eff = (1 + 0.10/365)^365 - 1
i_eff = (1 + 0.0002740)^365 - 1
i_eff = (1.0002740)^365 - 1 = 1.10516 - 1 = 0.10516
i_eff = 10.516% per year
```

Feature	Nominal Rate (r)	Effective Rate (i_eff)
Value	10% per year	10.516% per year
Compounding effect	Not included	Included
Used for	Stating the rate	Actual calculations
Comparison	Misleading for different frequencies	Valid for comparison

The difference between nominal (10%) and effective (10.516%) rates increases with compounding frequency. For daily compounding, the effective rate is 0.516% higher than the nominal rate.

Q15. Airplane ticket price will increase 8% in each of the next four years. The cost at the end of the first year will be Rs. 2,000. How much should be put away now to cover a passenger's travel home at the end of each year for the next four years? Assume i = 5%. [2080 Ashwin Q2b, 2078 Chaitra Q2b, 3 Marks]

Answer:

Given:

- Cash flows are a geometric gradient series (increasing by 8% per year)
- A1 = Rs. 2,000 at end of year 1, A2 = 2,000 × 1.08 = Rs. 2,160, A3 = Rs. 2,332.80, A4 = Rs. 2,519.42
- Growth rate g = 8% per year
- Interest rate i = 5% per year
- n = 4 years

Using Geometric Gradient PW Formula (g ≠ i):

```
P = A1 × [1 - (1+g)^n × (1+i)^(-n)] / (i - g)
P = 2,000 × [1 - (1.08)^4 × (1.05)^(-4)] / (0.05 - 0.08)
(1.08)^4 = 1.36049
(1.05)^4 = 1.21551
P = 2,000 × [1 - 1.36049/1.21551] / (-0.03)
P = 2,000 × [1 - 1.11930] / (-0.03)
P = 2,000 × [-0.11930] / (-0.03)
P = 2,000 × 3.9767 = Rs. 7,953.4
```

Amount to be set aside now: Rs. 7,953.40

Q16. What is the equal payment series for 10 years that is equivalent to a payment series of Rs. 12,000 at the end of the first year, decreasing by Rs. 1,000 each year over 10 years? Interest rate is 8% compounded annually. [2080 Ashwin Q2b, 4 Marks]

Answer:

Given:

- Payment at end of year 1: A1 = Rs. 12,000
- Annual decrease: G = -Rs. 1,000 (arithmetic gradient, decreasing)
- Number of years: n = 10
- Interest rate: i = 8% per year

The payment series: 12,000; 11,000; 10,000; ...; 3,000

Formula: A = A1 + G(A/G, i, n)

Where G = -1,000 (negative, decreasing gradient)

```
(A/G, i, n) = (1/i) - n/[(1+i)^n - 1]
(A/G, 8%, 10) = (1/0.08) - 10/[(1.08)^10 - 1]
(1.08)^10 = 2.15892
(A/G, 8%, 10) = 12.5 - 10/[2.15892 - 1] = 12.5 - 10/1.15892 = 12.5 - 8.6330 = 3.8670
```

$$A_equivalent = A1 + G \times (A/G) = 12{,}000 + (-1{,}000) \times 3.8670$$

$$A_equivalent = 12{,}000 - 3{,}867 = Rs.\ 8{,}133$$

The equivalent uniform annual payment series is Rs. 8,133 per year for 10 years.

Q17. If you deposited Rs. 5,00,000 now in a bank account which gives 6% interest per year, how many times would you be able to draw Rs. 10,000 per month with that money? [2080 Chaitra Q2b, 3 Marks]

Answer:

Given:

- P = Rs. 5,00,000 (present deposit)
- Nominal interest = 6% per year compounded yearly
- Monthly withdrawal = Rs. 10,000

Find: Number of months n

Step 1: Find effective monthly interest rate

$$i_monthly = 6\%/12 = 0.5\% \text{ per month} = 0.005$$

Step 2: Use P/A formula, solve for n

$$P = A \times (P/A, i, n)$$

$$5{,}00{,}000 = 10{,}000 \times [(1.005)^n - 1] / [0.005 \times (1.005)^n]$$

$$50 = [(1.005)^n - 1] / [0.005 \times (1.005)^n]$$

$$50 \times 0.005 \times (1.005)^n = (1.005)^n - 1$$

$$0.25 \times (1.005)^n = (1.005)^n - 1$$

$$1 = (1.005)^n - 0.25 \times (1.005)^n = 0.75 \times (1.005)^n$$

$$(1.005)^n = 1/0.75 = 1.3333$$

$$n \times \ln(1.005) = \ln(1.3333)$$

$$n = \ln(1.3333)/\ln(1.005) = 0.28768/0.004988 = 57.68 \text{ months}$$

You can draw Rs. 10,000 per month approximately 57 times (about 4 years and 9 months).

Q18. If you make equal monthly deposits of Rs. 5,000 into the bank for 10 years, saving accounts that pays interest rate of 6% compounded monthly, what would be the amount at the end of 15 years? [2074 Bhadra Q2a, 4 Marks]

Answer:

Given:

- Monthly deposit: A = Rs. 5,000
- Deposit period: 10 years (120 months)
- Interest rate: 6% compounded monthly => i = 6%/12 = 0.5% = 0.005 per month
- Find amount at end of 15 years (180 months)

Step 1: FV of deposits at end of 10 years (month 120)

$$FV_120 = 5{,}000 \times (F/A, 0.5\%, 120) = 5{,}000 \times [(1.005)^{120} - 1] / 0.005$$

$$(1.005)^{120} = 1.81940$$

$$FV_120 = 5{,}000 \times [1.81940 - 1] / 0.005 = 5{,}000 \times 163.879 = Rs.\ 8{,}19{,}397$$

Step 2: Advance FV_120 by 5 more years (60 months) at 0.5% per month

No new deposits from month 120 to 180, so only compounding continues:

$$FV_180 = 8{,}19{,}397 \times (1.005)^{60}$$

$$(1.005)^{60} = 1.34885$$

$$FV_180 = 8{,}19{,}397 \times 1.34885 = Rs.\ 11{,}05{,}960$$

Total amount at the end of 15 years: approximately Rs. 11,05,960.

Q19. How much rupees should you deposit now so that you will be able to draw Rs. 5,000 at the end of this month which increases by 2 percent per month for 15 years? Bank interest rate is 5% per year. [2074 Bhadra Q2b, 4 Marks]

Answer:

Given:

- First monthly withdrawal: A1 = Rs. 5,000 at end of month 1
- Growth rate: g = 2% per month
- Duration: 15 years = 180 months
- Nominal interest rate: 5% per year => i = 5%/12 = 0.4167% per month = 0.004167

Using Geometric Gradient PW Formula:

Since g = 2% per month >> i = 0.4167% per month (g ≠ i):

$$P = A1 \times [1 - (1+g)^n \times (1+i)^{(-n)}] / (i - g)$$

$$P = 5{,}000 \times [1 - (1.02)^{180} \times (1.004167)^{(-180)}] / (0.004167 - 0.02)$$

$$(1.02)^{180} = 35.3211$$

$$(1.004167)^{180} = 2.10718$$

$$P = 5{,}000 \times [1 - 35.3211/2.10718] / (-0.015833)$$

$$P = 5{,}000 \times [1 - 16.761] / (-0.015833)$$

$$P = 5{,}000 \times [-15.761] / (-0.015833)$$

$$P = 5{,}000 \times 995.47 = Rs.\ 49{,}77{,}350$$

Amount to be deposited now: approximately Rs. 49,77,350.

Q20. How much money must be invested in an account that pays 9% per year interest compounded quarterly to draw Rs. 6,000 at the end of first year and drawing amount increases by Rs. 1,000 thereafter up to 10 years (use gradient formula)? [2079 Jestha Q2, 8 Marks]

Answer:

Given:

- Withdrawals: Rs. 6,000 at end of year 1, Rs. 7,000 at end of year 2, ..., Rs. 15,000 at end of year 10
- This is an arithmetic gradient series: A1 = Rs. 6,000, G = Rs. 1,000
- Nominal interest rate = 9% per year compounded quarterly
- n = 10 years

Step 1: Find effective annual interest rate

```
i_eff = (1 + 0.09/4)^4 - 1 = (1.0225)^4 - 1 = 9.308% per year
```

Use i = 9.308% for annual gradient calculations.

Step 2: Decompose into uniform series A1 and gradient G

Total PW = PW of uniform part + PW of gradient part

```
PW = A1(P/A, i, n) + G(P/G, i, n)
```

Step 3: Calculate (P/A, 9.308%, 10)

```
(P/A, 9.308%, 10) = [(1.09308)^10 - 1] / [0.09308 × (1.09308)^10]
(1.09308)^10 = 2.43475
(P/A) = [2.43475 - 1] / [0.09308 × 2.43475] = 1.43475 / 0.22657 = 6.3327
```

Step 4: Calculate (P/G, 9.308%, 10)

```
(P/G, 9.308%, 10) = [(1.09308)^10 - (9.308%×10) - 1] / [(9.308%)^2 ×
(1.09308)^10]
= [2.43475 - 0.9308 - 1] / [(0.09308)^2 × 2.43475]
= 0.50395 / [0.008664 × 2.43475] = 0.50395 / 0.02108 = 23.904
```

Step 5: Calculate PW

```
PW = 6,000 × 6.3327 + 1,000 × 23.904
PW = 37,996.2 + 23,904 = Rs. 61,900.2
```

Amount to be invested: approximately Rs. 61,900.

Q21. What will be the amount at the end of 10 years if you deposit Rs. 5,000 per month for five years continuously if nominal interest rate is 10% compounded quarterly? [2071 Magh Q2a, 3 Marks]

Answer:

Given:

- Monthly deposit: A = Rs. 5,000
- Deposit period: 5 years = 60 months
- Nominal interest rate: r = 10% compounded quarterly
- Effective monthly rate: i_m = (1 + 0.10/4)^(1/3) - 1 = (1.025)^(1/3) - 1 = 0.008265 = 0.8265%/month
- Total period: 10 years (deposits for first 5 years, then left to grow for next 5 years)

Step 1: FV of 60 monthly deposits at end of year 5

```
FV_5 = 5,000 × [(1.008265)^60 - 1] / 0.008265
(1.008265)^60 = (1.025)^20 = 1.63862
FV_5 = 5,000 × [1.63862 - 1] / 0.008265 = 5,000 × 77.30 = Rs. 3,86,500
```

Step 2: Advance FV_5 by another 5 years (no deposits) at effective annual rate

Effective annual rate = (1.025)^4 - 1 = 10.381%

```
FV_10 = 3,86,500 × (1.10381)^5 = 3,86,500 × 1.63862 = Rs. 6,33,325
```

Amount at end of 10 years: approximately Rs. 6,33,325.

Q22. Kamal aged 30 years now had borrowed Rs. 5,00,000 from a bank 5 years ago. The bank charged interest rate at 10% per year compounded monthly. What will be the semiannual equal installment to clear the loan if his payments start now and he wishes to repay within 5 years from now? [2078 Poush Q2a, 5 Marks]

Answer:

Given:

- Original loan: Rs. 5,00,000 borrowed 5 years ago
- Interest rate: 10% per year compounded monthly => i = 10%/12 = 0.8333%/month
- Loan has been outstanding for 5 years (60 months) — need to find current outstanding balance
- Repayment: semiannual payments over 5 years from now (10 semiannual payments)

Step 1: Find current outstanding balance (after 5 years of accumulation, no payments yet)

```
Outstanding = 5,00,000 × (1 + 0.10/12)^(12×5)
Outstanding = 5,00,000 × (1.008333)^60
(1.008333)^60 = 1.64534
Outstanding = 5,00,000 × 1.64534 = Rs. 8,22,670
```

Step 2: Find effective semiannual interest rate

```
i_semi = (1 + 0.10/12)^6 - 1 = (1.008333)^6 - 1 = 1.05115 - 1 = 5.115%
```

Step 3: Find semiannual payment A

n = 10 semiannual periods, P = Rs. 8,22,670

```
A = P × (A/P, i_semi, n) = P × i_semi(1+i_semi)^10 / [(1+i_semi)^10 - 1]
(1.05115)^10 = 1.65012
A = 8,22,670 × [0.05115 × 1.65012] / [1.65012 - 1]
A = 8,22,670 × [0.08440] / [0.65012]
A = 8,22,670 × 0.12982 = Rs. 1,06,801
```

Semiannual payment to clear the loan in 5 years: approximately Rs. 1,06,801.

Q23. A person invests a sum of Rs. 50,000 in a bank at a nominal interest rate of 18% per year. The compounding is monthly. Find the maturity amount of the deposit after 15 years. Also briefly explain the importance of time value of money. [2073 Magh Q2, 3+3 Marks]

Answer:

Given: P = Rs. 50,000, r = 18% per year, m = 12 (monthly compounding), n = 15 years

Maturity Amount Calculation:

```
F = P × (1 + r/m)^(m×n) = 50,000 × (1 + 0.18/12)^(12×15)
F = 50,000 × (1.015)^180
(1.015)^180 = 14.1300
F = 50,000 × 14.1300 = Rs. 7,06,500
```

Maturity amount after 15 years: Rs. 7,06,500.

Importance of Time Value of Money:

1. Investment Appraisal: TVM allows engineers to correctly evaluate whether an investment will generate enough future returns to justify the present expenditure.
2. Loan Analysis: Understanding TVM enables correct calculation of loan repayments, interest costs, and amortization schedules.
3. Economic Comparison: TVM enables proper comparison of cash flows occurring at different points in time by converting them to a common reference (present, future, or annual worth).
4. Retirement Planning: TVM calculations help in planning for future financial needs by determining how much to save today.
5. Inflation Adjustment: TVM principles are the foundation for inflation analysis in engineering economics.

Q24. If you planned to borrow some amount of money from your friend right now by promising that you will repay her at the end of this year by Rs. 10,000 and will be decreased the amount for succeeding years by Rs. 1,000 per year then after for another next 4 years. Consider 12% interest pa compounding daily, what amount will she be ready to lend you? [2079 Ashwin Q2b, 4 Marks]

Answer:

Given:

- Cash flows: Rs. 10,000 at end of year 1, Rs. 9,000 at end of year 2, Rs. 8,000 at year 3, Rs. 7,000 at year 4, Rs. 6,000 at year 5
- This is a decreasing arithmetic gradient: A1 = 10,000, G = -1,000, n = 5 years
- Nominal interest: 12% compounded daily

Step 1: Find effective annual interest rate (daily compounding)

$$i_eff = (1 + 0.12/365)^{365} - 1 = (1.000329)^{365} - 1 = 1.12747 - 1 = 12.747\%$$

Step 2: Calculate PW with A1 = 10,000, G = -1,000, i = 12.747%, n = 5

$$PW = A1(P/A, i, n) + G(P/G, i, n)$$

$$(P/A, 12.747\%, 5) = [(1.12747)^5 - 1] / [0.12747 \times (1.12747)^5]$$

$$(1.12747)^5 = 1.81483$$

$$(P/A) = [0.81483] / [0.12747 \times 1.81483] = 0.81483 / 0.23124 = 3.5237$$

$$(P/G, 12.747\%, 5) = [(1.12747)^5 - 12.747\%(5) - 1] / [(0.12747)^2 \times (1.12747)^5]$$

$$= [1.81483 - 0.63735 - 1] / [0.016249 \times 1.81483]$$

$$= 0.17748 / 0.02949 = 6.018$$

$$PW = 10{,}000 \times 3.5237 + (-1{,}000) \times 6.018$$

$$PW = 35{,}237 - 6{,}018 = \text{Rs. } 29{,}219$$

Your friend will be ready to lend approximately Rs. 29,219.

Q25. A man in planning to retire in 25 years. He wishes to deposit regular money every month until he retires so that he will receive annual payments of Rs. 4,50,000 after the first year of his retirement for the next 10 years. How

much he should deposit if the interest rate is 8%, compounded monthly? [2076 Baisakh Q2b, 5 Marks]

Answer:

Given:

- Annual withdrawal after retirement: Rs. 4,50,000 per year for 10 years
- First withdrawal: 1 year after retirement (i.e., at year 26 from now, year 1 of retirement)
- Monthly deposit for 25 years
- Interest rate: 8% compounded monthly => i = 8%/12 = 0.6667%/month

Step 1: Effective annual interest rate

```
i_eff = (1 + 0.08/12)^12 - 1 = (1.00667)^12 - 1 = 1.08300 - 1 = 8.3% per year
```

Step 2: PW of retirement annuity at moment of retirement (end of year 25)

```
PW_retire = 4,50,000 × (P/A, 8.3%, 10)
(P/A, 8.3%, 10) = [(1.083)^10 - 1] / [0.083 × (1.083)^10]
(1.083)^10 = 2.22261
(P/A) = [2.22261 - 1] / [0.083 × 2.22261] = 1.22261 / 0.18448 = 6.6274
PW_retire = 4,50,000 × 6.6274 = Rs. 29,82,330
```

Step 3: Find monthly deposit A to accumulate Rs. 29,82,330 in 25 years

```
FV = A × (F/A, 0.6667%, 300)
(F/A, 0.6667%, 300) = [(1.00667)^300 - 1] / 0.00667
(1.00667)^300 = (1.083)^25 = 7.1226
(F/A) = [7.1226 - 1] / 0.00667 = 6.1226 / 0.00667 = 918.1
29,82,330 = A × 918.1
A = 29,82,330 / 918.1 = Rs. 3,248.6 per month
```

Monthly deposit required: approximately Rs. 3,249 per month for 25 years.

Q26. What is the nominal and effective interest rate? Evaluate FW at the end of 10 years with 12% interest rate compounded monthly of a cash flow of Rs. 40,000 at the beginning of each year for 5 years. [2073 Bhadra Q2, 2+4 Marks]

Answer:

Nominal vs. Effective Interest Rate

Nominal Interest Rate: The stated annual rate without considering intra-year compounding effects. Symbol: r.

Effective Interest Rate: The true annual rate after accounting for compounding within the year. Symbol: i_eff = (1 + r/m)^m - 1

Calculation: r = 12% compounded monthly

Given:

- A = Rs. 40,000 paid at the beginning of each year (annuity due) for 5 years
- Interest: 12% compounded monthly

- Find: Future Worth at end of year 10

Step 1: Effective annual interest rate

```
i_eff = (1 + 0.12/12)^12 - 1 = (1.01)^12 - 1 = 1.12683 - 1 = 12.683%
```

Step 2: FW of annuity due at end of year 5

Since payments are at beginning of each year (annuity due), multiply standard FW by (1+i):

```
FW_5 = A × (F/A, i, n) × (1+i)    [annuity due]
(F/A, 12.683%, 5) = [(1.12683)^5 - 1] / 0.12683
(1.12683)^5 = 1.81596
(F/A) = [1.81596 - 1] / 0.12683 = 0.81596 / 0.12683 = 6.4336
FW_5 = 40,000 × 6.4336 × 1.12683 = 40,000 × 7.2479 = Rs. 2,89,916
```

Step 3: Advance FW_5 by another 5 years (no deposits, years 6-10)

```
FW_10 = FW_5 × (1.12683)^5 = 2,89,916 × 1.81596 = Rs. 5,26,567
```

Future Worth at end of 10 years: approximately Rs. 5,26,567.

Q27. If you wish to withdraw Rs. 2,000 at the end of 1st year and expecting to be increased by 15% p.a. after till end of 8 years, what amount needs to be deposited in a bank right now which has an interest of 15% p.a.? [2075 Bhadra Q2b, 4 Marks]

Answer:

Given:

- Geometric gradient withdrawals: A1 = Rs. 2,000 at end of year 1, increasing 15% per year
- Growth rate g = 15% per year
- Interest rate i = 15% per year
- n = 8 years

Note: g = i = 15%, so we use the special case formula:

Special Case Formula when g = i:

```
P = A1 × n / (1 + i) = 2,000 × 8 / 1.15
P = 16,000 / 1.15 = Rs. 13,913.04
```

Amount to be deposited now: Rs. 13,913.04

Q28. If you deposit Rs. 10,000 in a saving account now which gives 10% nominal interest rate, what will be the amount after 5 years if interest is compounded (i) semi-annually (ii) monthly? [2070 Magh Q2a, 2 Marks]

Answer:

Given: P = Rs. 10,000, r = 10%, n = 5 years

(i) Semi-annual compounding (m = 2):

```
F = P × (1 + r/m)^(m×n) = 10,000 × (1 + 0.10/2)^(2×5)
F = 10,000 × (1.05)^10 = 10,000 × 1.62889 = Rs. 16,289
```

(ii) Monthly compounding (m = 12):

$$F = 10{,}000 \times (1 + 0.10/12)^{(12\times5)} = 10{,}000 \times (1.008333)^{60}$$

$$(1.008333)^{60} = 1.64533$$

$$F = 10{,}000 \times 1.64533 = \text{Rs. } 16{,}453$$

Compounding	Formula	Future Amount
Semi-annual (m=2)	10,000 × (1.05)^10	Rs. 16,289
Monthly (m=12)	10,000 × (1.00833)^60	Rs. 16,453

Observation: Higher compounding frequency (monthly) yields a higher amount (Rs. 16,453) compared to semi-annual (Rs. 16,289).

Q29. What will be the amount at the end of 10 years if you deposit Rs. 5,000 per month for five years continuously if the nominal interest rate is 10% compounded quarterly for the first five years, after that interest rate changes to 6% per year? [2077 Chaitra, 1+4 Marks]

Answer:

Given:

- Monthly deposit: A = Rs. 5,000 for years 1 to 5 (60 months)
- Interest rate years 1-5: r = 10% compounded quarterly => effective monthly rate: i_1 = (1.025)^(1/3) - 1 = 0.8265%
- Interest rate years 6-10: 6% per year = 0.5% per month

Step 1: FV at end of year 5 (during 10% quarterly period)

$$FV_5 = 5{,}000 \times (F/A,\ 0.8265\%,\ 60) = 5{,}000 \times [(1.008265)^{60} - 1] / 0.008265$$

$$(1.008265)^{60} = (1.025)^{20} = 1.63862$$

$$FV_5 = 5{,}000 \times [0.63862 / 0.008265] = 5{,}000 \times 77.266 = \text{Rs. } 3{,}86{,}330$$

Step 2: Advance FV_5 from year 5 to year 10 at 6% annual (0.5% monthly)

$$FV_10 = 3{,}86{,}330 \times (1.005)^{60} = 3{,}86{,}330 \times 1.34885 = \text{Rs. } 5{,}21{,}198$$

Total amount at end of 10 years: approximately Rs. 5,21,198.

Q30. A couple is planning for their child's education. They wish to deposit Rs. 10,000 now in a bank account that gives 12% per year compounded monthly and increase the amount by Rs. 2,000 each year from the previous year for next 9 years. How much amount they will expect at the end of 10 years? [2077 Chaitra, 5 Marks]

Answer:

Given:

- Deposit at t=0: Rs. 10,000
- Deposit at t=1: Rs. 12,000 (increases by Rs. 2,000 each year)
- Deposits: 10,000 at t=0, 12,000 at t=1, 14,000 at t=2, ..., 28,000 at t=9
- Arithmetic gradient: A0 = 10,000, G = 2,000
- Interest: 12% per year compounded monthly => effective annual rate = (1.01)^12 - 1 = 12.683%

Step 1: Find FV of each deposit at end of year 10

Instead of PW approach, compute FV directly:

Deposit at t=k earns for (10-k) years at 12.683%

```
FV = Sum[A_k × (1.12683)^(10-k)]  for k = 0 to 9
```

A_k = 10,000 + 2,000×k

Calculating each term:

Year k	Deposit A_k	(1.12683)^(10-k)	FV
0	10,000	(1.12683)^10 = 3.2745	32,745
1	12,000	(1.12683)^9 = 2.9057	34,868
2	14,000	(1.12683)^8 = 2.5784	36,098
3	16,000	(1.12683)^7 = 2.2880	36,608
4	18,000	(1.12683)^6 = 2.0305	36,549
5	20,000	(1.12683)^5 = 1.8022	36,044
6	22,000	(1.12683)^4 = 1.5993	35,185
7	24,000	(1.12683)^3 = 1.4194	34,066
8	26,000	(1.12683)^2 = 1.2597	32,752
9	28,000	(1.12683)^1 = 1.1268	31,550

```
Total FV = 32,745 + 34,868 + 36,098 + 36,608 + 36,549 + 36,044 + 35,185 +
34,066 + 32,752 + 31,550

Total FV = Rs. 3,46,465
```

Expected amount at the end of 10 years: approximately Rs. 3,46,465.

Q31. Find the value of P if i = 10%. Given cash flow: Rs. 10 at t=0, Rs. 12 at t=1, Rs. 14 at t=2, Rs. 16 at t=3, Rs. 16 at t=4, Rs. 16 at t=5, Rs. 16 at t=6 (amounts in thousands). Use gradient formula also. [2070 Magh Q2b, 4 Marks]

Answer:

Given cash flow description (from exam diagram): The cash flows increase from Rs. 10 to Rs. 16 (in thousands) and then remain constant. This involves both arithmetic gradient and uniform series components.

Cash flows (in Rs.): t=1: 10,000; t=2: 12,000; t=3: 14,000; t=4: 16,000; t=5: 16,000; t=6: 16,000 (with P = ? at t=0)

Wait — the gradient part is: 10, 12, 14, 16 for years 1-4 (G = 2,000), then uniform 16,000 for years 5-6.

Approach 1: Direct PW calculation of each cash flow

```
P = 10,000(P/F,10%,1) + 12,000(P/F,10%,2) + 14,000(P/F,10%,3) +
16,000(P/F,10%,4)

+ 16,000(P/F,10%,5) + 16,000(P/F,10%,6)

(P/F factors): 0.9091, 0.8264, 0.7513, 0.6830, 0.6209, 0.5645

P = 10,000(0.9091) + 12,000(0.8264) + 14,000(0.7513) + 16,000(0.6830)
```

```
+ 16,000(0.6209) + 16,000(0.5645)
P = 9,091 + 9,917 + 10,518 + 10,928 + 9,934 + 9,032 = Rs. 59,420
```

Approach 2: Using Gradient Formula

First 4 payments: A1 = 10,000, G = 2,000, n = 4, i = 10%

```
PW_gradient = 10,000(P/A,10%,4) + 2,000(P/G,10%,4)
(P/A,10%,4) = 3.1699,  (P/G,10%,4) = 4.3781
PW_gradient = 10,000(3.1699) + 2,000(4.3781) = 31,699 + 8,756 = Rs. 40,455
```

Payments at years 5 and 6 (Rs. 16,000 each), discounted back to t=0:

```
PW_56 = 16,000(P/F,10%,5) + 16,000(P/F,10%,6) = 16,000(0.6209 + 0.5645) =
16,000(1.1854) = Rs. 18,966
P = PW_gradient + PW_56 = 40,455 + 18,966 = Rs. 59,421
```

P = Rs. 59,420 (approximately Rs. 59,421).

Q32. If cash flow at the end of 1st year is Rs. 5,000; find FW using gradient formula taking MARR = 7% when Cash flow increases by 10% over 5 years. [2078 Baishakh Q4, 2 Marks]

Answer:

Given:

- A1 = Rs. 5,000 at end of year 1
- Geometric gradient: g = 10% per year
- i = MARR = 7% per year
- n = 5 years

Cash flows: 5,000; 5,500; 6,050; 6,655; 7,320.50

Step 1: Find PW using geometric gradient formula (g ≠ i)

```
PW = A1 × [1 - (1+g)^n(1+i)^(-n)] / (i - g)
PW = 5,000 × [1 - (1.10)^5(1.07)^(-5)] / (0.07 - 0.10)
(1.10)^5 = 1.61051,   (1.07)^5 = 1.40255
PW = 5,000 × [1 - 1.61051/1.40255] / (-0.03)
PW = 5,000 × [1 - 1.14832] / (-0.03)
PW = 5,000 × [-0.14832] / (-0.03) = 5,000 × 4.944 = Rs. 24,720
```

Step 2: Find FW at end of year 5

```
FW = PW × (F/P, 7%, 5) = 24,720 × (1.07)^5 = 24,720 × 1.40255 = Rs. 34,671
```

Future Worth at end of year 5: Rs. 34,671.

Q33. How much should you deposit at present for a monthly draw of Rs. 15,000 throughout 4 years when interest rate is 6% per year? [2078 Baishakh Q3, 4 Marks]

Answer:

Given:

- Monthly withdrawal: A = Rs. 15,000

- Duration: 4 years = 48 months
- Annual interest rate: 6% per year => monthly rate: i = 6%/12 = 0.5% = 0.005

Using PW formula for uniform series:

$$P = A \times (P/A, i, n) = 15{,}000 \times [(1.005)^{48} - 1] / [0.005 \times (1.005)^{48}]$$

$$(1.005)^{48} = 1.27049$$

$$P = 15{,}000 \times [1.27049 - 1] / [0.005 \times 1.27049]$$

$$P = 15{,}000 \times 0.27049 / 0.0063525$$

$$P = 15{,}000 \times 42.58 = \text{Rs. } 6{,}38{,}700$$

Amount to be deposited now: Rs. 6,38,700.

Q34. Suppose that you make a deposit of Rs. 5,000 per month in a saving account which gives 12% interest compounded quarterly for the first three years and 9% compounded monthly for the last two years. What amount do you expect at the end of 5 years? [2077 Poush Q2a, 3 Marks]

Answer:

Given:

- Monthly deposit: A = Rs. 5,000
- Years 1-3 (36 months): r1 = 12% quarterly => i1_monthly = (1.03)^(1/3) - 1 = 0.9902%
- Years 4-5 (24 months): r2 = 9% monthly => i2 = 9%/12 = 0.75%

Step 1: FV at end of year 3 (from monthly deposits at 0.9902% monthly)

$$FV_3 = 5{,}000 \times [(1.009902)^{36} - 1] / 0.009902$$

$$(1.009902)^{36} = (1.03)^{12} = 1.42576$$

$$FV_3 = 5{,}000 \times [1.42576 - 1] / 0.009902 = 5{,}000 \times 43.00 = \text{Rs. } 2{,}15{,}000$$

Step 2: Advance FV_3 to end of year 5 AND add deposits from years 4-5

Carry forward FV_3 for 24 months at 0.75% monthly:

$$FV_3_carried = 2{,}15{,}000 \times (1.0075)^{24} = 2{,}15{,}000 \times 1.19641 = \text{Rs. } 2{,}57{,}228$$

FV of 24 monthly deposits (years 4-5) at 0.75% monthly:

$$FV_new = 5{,}000 \times [(1.0075)^{24} - 1] / 0.0075 = 5{,}000 \times 25.932 = \text{Rs. } 1{,}29{,}660$$

$$Total\ FV_5 = 2{,}57{,}228 + 1{,}29{,}660 = \text{Rs. } 3{,}86{,}888$$

Expected amount at end of 5 years: approximately Rs. 3,86,888.

--- End of Chapter 2 Questions & Answers ---